Fitting the frog load model to a field trial

What kept Bd low on treated frogs in Dusy Basin?

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The trial

In July 2012, Knapp et al. (2022) treated subadult Sierra Nevada yellow-legged frogs (*Rana sierrae*) in Dusy Basin, California, during a Bd epizootic:

- days −7 to −1: a daily bath in the antifungal itraconazole;
- days 0 and 1: a bath in a culture of *Janthinobacterium lividum*, a skin bacterium that inhibits Bd, isolated from a frog in the same basin;
- release into the pond, then swabs on days 12, 37 and 56.

Untreated wild frogs and a group of control frogs (marked, never bathed) were swabbed at the same visits. Every swab was tested for Bd (ITS copies) and for *J. lividum* (genomic equivalents). Frogs were marked by group, not individually, so each visit samples the group, not the same animals.

This vignette fits the frog load equation of the package to these swabs. It asks why Bd stayed low on treated frogs for weeks, and whether *J. lividum* had anything to do with it.

> **!** Important
>
> The load equation and the fits are new work in this package, not a published analysis. The data are from Knapp et al. (2022), CC BY 4.0, bundled in `data/knapp2022`. The within-frog rate `r0` comes from the Briggs et al. (2010) parameters in the package's compendium; everything else is fitted here.

```
using Chytridiomycosis, Plots, Printf
using Distributions: median
include("../diagram.jl")
default(linewidth = 2, size = (720, 380), legendfontsize = 8, guidefontsize
 ↪  = 9,
    titlefontsize = 10, left_margin = 4Plots.mm, bottom_margin = 4Plots.mm)

swabs = load_knapp2022()
groups = (:treated, :wild, :control)
days = [-7, 0, 12, 37, 56]
nothing
```

The data

```
println("| Group | Day | Swabs | No Bd | Median Bd load | Median *J.
 ↪  lividum* |")
println("|---|---|---|---|---|---|")
for g in groups, d in days
    s = filter(x -> x.group === g && x.day == d, swabs)
    isempty(s) && continue
    bd = [x.bd for x in s if !isnan(x.bd)]
    jl = [x.jl for x in s if !isnan(x.jl)]
    @printf("| %s | %d | %d | %d | %s | %s |\n", g, d, length(s),
     ↪  count(==(0), bd),
        isempty(bd) ? "–" : @sprintf("%.3g", median(bd)),
        isempty(jl) ? "–" : @sprintf("%.3g", median(jl)))
end
```

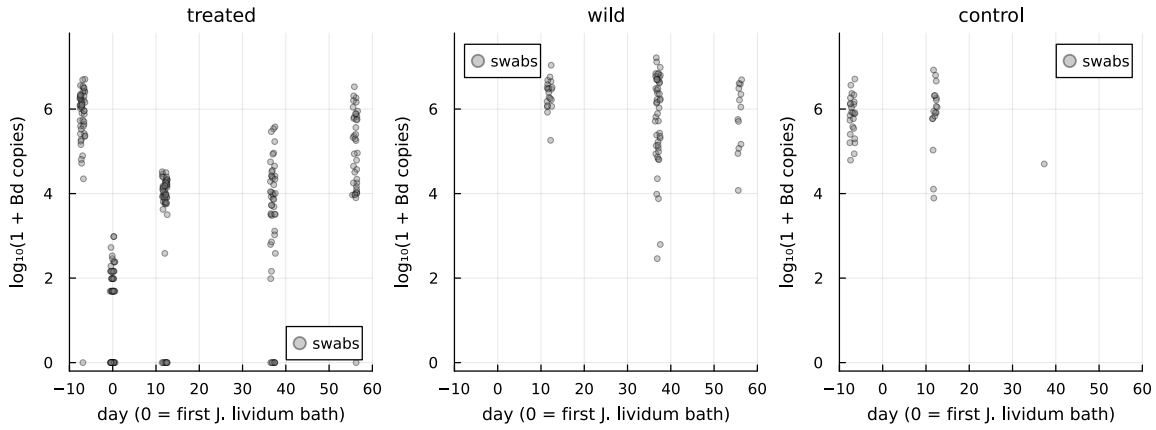
Group	Day	Swabs	No Bd	Median Bd load	Median *J. lividum*
treated	-7	51	1	1.13e+06	–
treated	0	52	20	48	0

Group	Day	Swabs	No Bd	Median Bd load	Median *J. lividum*
treated	12	57	9	9.65e+03	155
treated	37	47	7	7.44e+03	2
treated	56	39	1	1.8e+05	0
wild	12	20	0	2.72e+06	50.5
wild	37	48	0	1.42e+06	4
wild	56	13	0	1.12e+06	0
control	-7	25	0	8.16e+05	–
control	12	18	0	9.96e+05	1.53e+03
control	37	1	0	5.04e+04	158

Three things stand out:

- Treated frogs stayed low for weeks. Itraconazole took their loads from about a million copies to about 50. Twelve days after release they carried about 10,000, and still did on day 37, while wild frogs carried more than a million. Only by day 56 had they risen again.
- J. lividum faded on everyone. On treated frogs it fell from about 150 genomic equivalents on day 12 to almost none by day 56. It fell at a similar rate on wild frogs, which were never bathed.
- Controls carried more J. lividum than treated frogs (about 1,500 on day 12), and their loads were as high as those of wild frogs.

```
using Random: Xoshiro
function data_panel(g; title = string(g))
    s = filter(x -> x.group === g && !isnan(x.bd), swabs)
    jitter = 1.2 .* (rand(Xoshiro(1), length(s)) .- 0.5)
    scatter([x.day for x in s] .+ jitter, log10.(1 .+ [x.bd for x in s]); ms
     ↪  = 2.5, alpha = 0.4,
        color = :grey, label = "swabs", title = title, xlabel = "day (0 =
 ↪  first J. lividum bath)",
        ylabel = "log₁₀(1 + Bd copies)", ylim = (-0.2, 7.8), xlim = (-10,
 ↪  60))
end
plot(data_panel.(groups)...; layout = (1, 3), size = (900, 330))
```



The model

Each group is described by its median load L (ITS copies per swab):

$$\frac{dL}{dt} = \iota + r_0 L - s(t)\,\frac{L}{h + L}.$$

- ι is infection from outside the frog, by zoospores in the pond.
- $r_0 = \eta\nu f - \sigma$ is the within-frog rate of the Briggs et al. (2010) model. It is negative, so a frog's own zoospores cannot sustain its load.
- $s(t)\,L/(h + L)$ is extra clearance. It is strongest when the load is small and saturates when the load is large.

Without extra clearance, every frog settles at $L^* = \iota/(-r_0)$, with a time scale of $1/|r_0| \approx 6$ days. That is the problem: on its own the equation would return treated frogs to the wild level within two weeks of release.

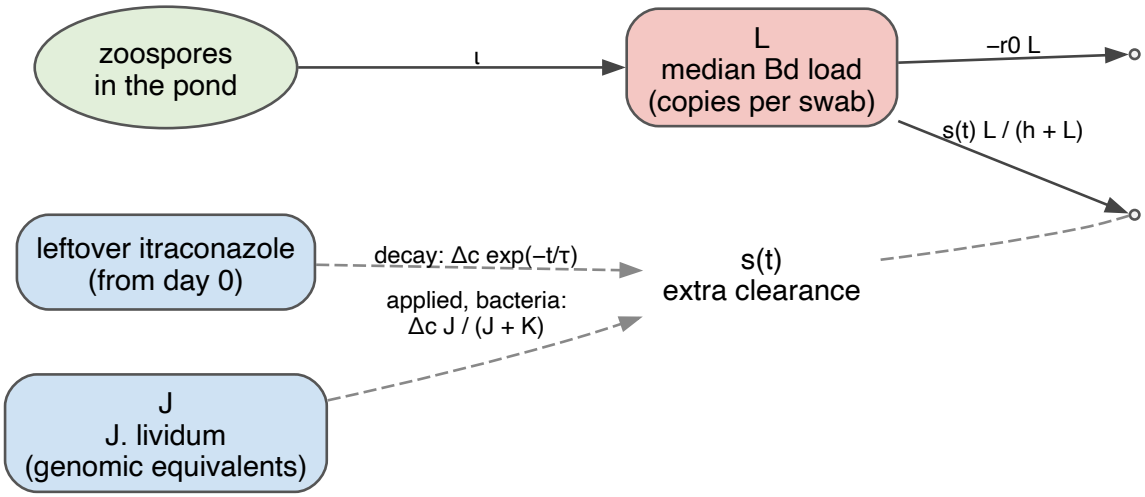


Figure 1: The fitted model for one group of frogs. Solid arrows are flows of Bd load; dashed arrows show what sets the extra clearance under each mechanism.

The four mechanisms differ only in the extra clearance $s_g(t)$ of group g (treated T, wild W or control C), with time t in days from the first bath:

$$s_g(t) = \begin{cases} 0 & \text{none} \\ \Delta c\, e^{-t/\tau} \text{ for } g = T,\ t \geq 0, \text{ else } 0 & \text{decay} \\ \Delta c\, J_T(t)/(J_T(t) + K) \text{ for } g = T,\ t \geq 0, \text{ else } 0 & \text{applied} \\ \Delta c\, J_g(t)/(J_g(t) + K) & \text{bacteria} \end{cases}$$

J. lividum falls exponentially from its day-12 level a_g, with half-life h_T on treated frogs and h_W on wild and control frogs, which are held at their day-12 level before day 12:

$$J_T(t) = a_T\, 2^{-(t-12)/h_T}, \qquad J_g(t) = a_g\, 2^{-\max(t-12,\,0)/h_W} \quad (g = W, C).$$

Untreated groups start at their equilibrium, the positive root of $\iota + r_0 L - sL/(h + L) = 0$. Treated frogs start at a fitted load L_0 on day 0; their swabs on day -7, before treatment, are predicted by the untreated equilibrium (for `bacteria`, with the controls' *J. lividum*). Each swab y is lognormal around the median, and a swab with nothing detected is counted as below one copy:

$$\log y \sim \text{Normal}(\log L_g(t),\ \sigma^2), \qquad \Pr(y = 0) = \Pr(\log y < 0),$$

and likewise for *J. lividum* with its own standard deviation σ_J.

Symbol	Meaning	Value
ι	infection from the pond (copies per swab per day)	fitted
$r_0 = \eta\nu f - \sigma$	within-frog rate (per day)	compendium (briggs2010)
Δc, h	most extra clearance (copies per day); load at which it halves	fitted
τ	time scale of waning, `decay` (days)	fitted
K	*J. lividum* giving half the extra clearance	fitted
L_0	load of treated frogs on day 0	fitted
$a_T, a_W, a_C; h_T, h_W$	*J. lividum* on day 12; its half-lives (days)	fitted
σ, σ_J	standard deviations of log Bd and log *J. lividum*	fitted

```
mechanisms = (:none, :decay, :applied, :bacteria)
models = Dict(mech => KnappLoadModel(mech) for mech in mechanisms)
fits = Dict(mech => maximum_likelihood(models[mech]) for mech in mechanisms)
r0 = models[:none].r0
@printf("r0 = %.4f per day\n", r0)
```

```
r0 = -0.1562 per day
```

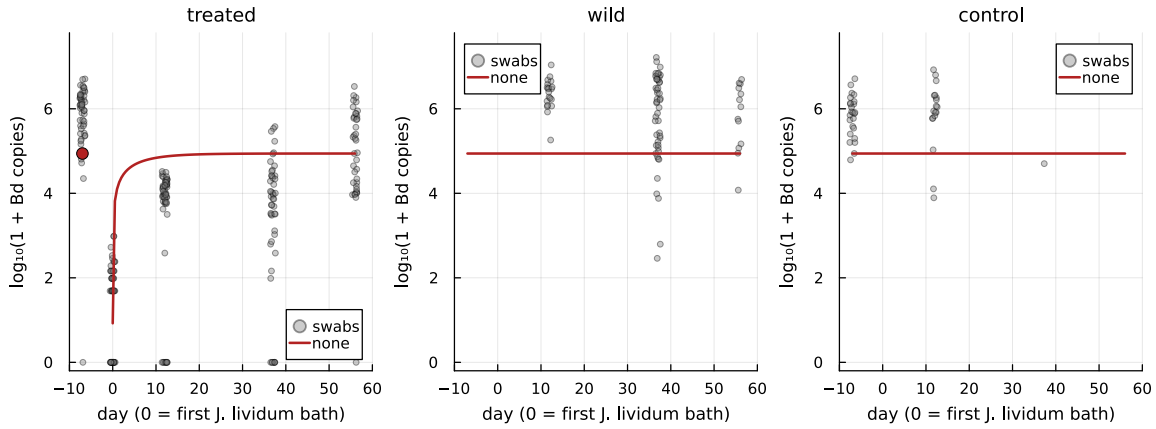
The fits below compare four explanations for $s(t)$, from simplest to most complex. Each is fitted to every swab, Bd and *J. lividum* alike, from eight starting points. In the figures, the red line is the fitted median load. Swabs with no Bd detected are plotted at 0 and count in the fit, so the median for treated frogs lies below the cloud of positive swabs.

```
fine = vcat(-7.0, 0.0:0.5:56.0)
function fit_panels(mech; label = string(mech))
    pred = predicted_loads(models[mech], fits[mech].x; days = fine)
    panels = map(groups) do g
        p = data_panel(g)
        y = getfield(pred, g).bd
        keep = g === :treated ? (fine .>= 0) : trues(length(fine))
        plot!(p, fine[keep], log10.(1 .+ y[keep]); color = :firebrick, label
         ↪  = label)
        g === :treated && scatter!(p, [-7], [log10(1 + y[1])]; color =
 ↪  :firebrick, ms = 5,
            label = "")
        p
```

```
    end
    plot(panels...; layout = (1, 3), size = (900, 330))
end
nothing
```

Step 1: no extra clearance

```
fit_panels(:none)
```

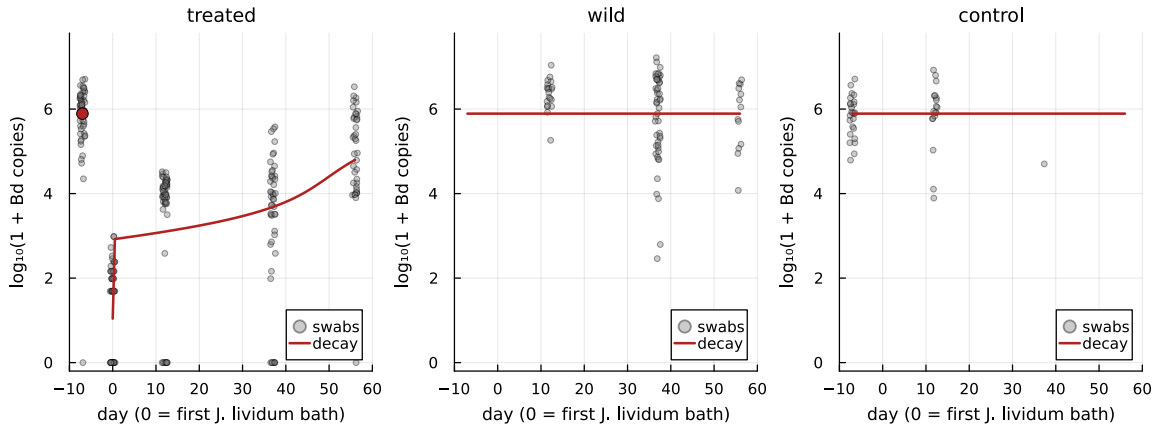


Treated frogs return to the common level by day 12 and stay there. The fit compromises: it sets that level between treated and untreated frogs, too low for wild frogs and too high for treated ones.

Step 2: suppression that wanes

Now treated frogs get extra clearance $s(t) = \Delta c\, e^{-t/\tau}$ from day 0 on, as leftover antifungal drug would provide. Untreated frogs get none.

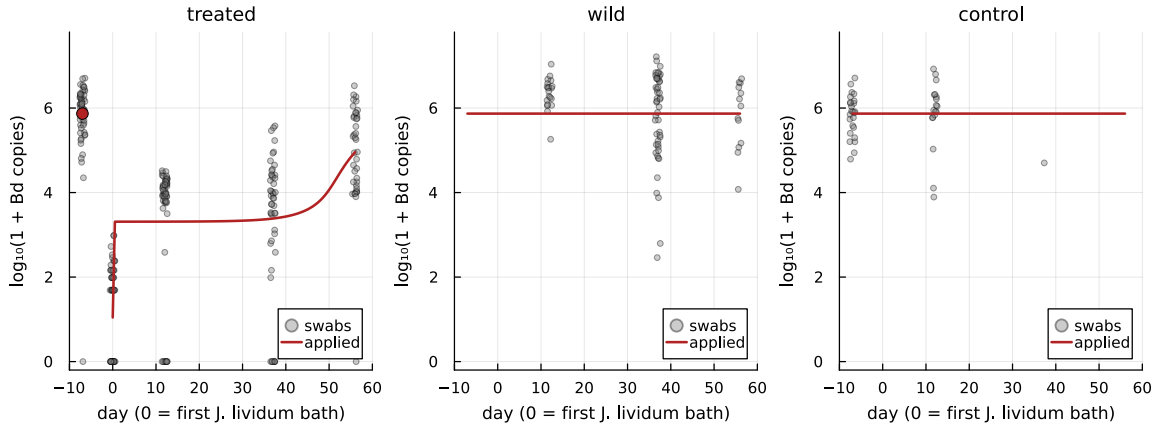
```
fit_panels(:decay)
```



Step 3: suppression from the applied bacteria

The extra clearance on treated frogs now follows their *J. lividum*: $s(t) = \Delta c\, J(t)/(J(t) + K)$. It is strong while the bacteria are abundant and fades as they disappear. Resident *J. lividum* on wild and control frogs is assumed to have no effect, as if the applied isolate differed from the residents (the assay cannot tell them apart).

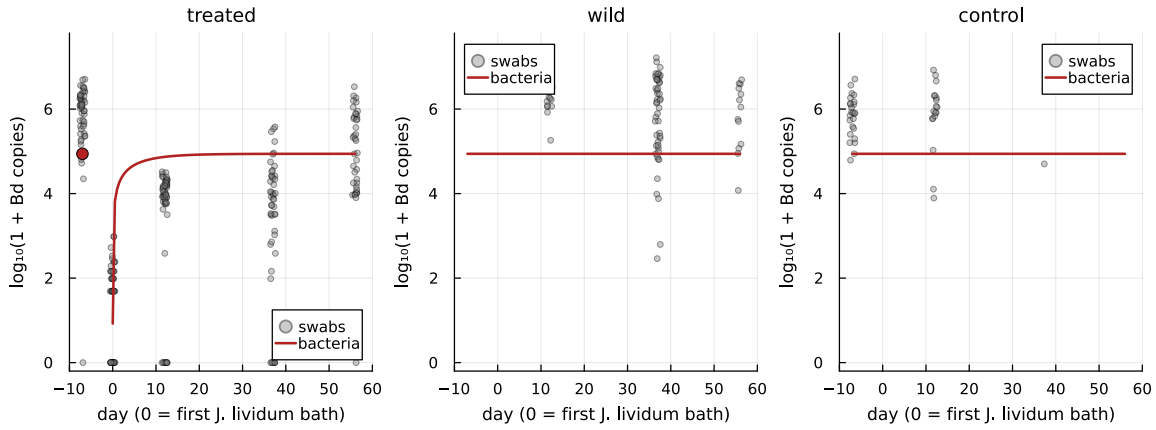
```
fit_panels(:applied)
```



Step 4: every J. lividum protects

The same relation now applies to every frog, using each group's own *J. lividum*. If the species protects in proportion to its abundance, the residents on wild and control frogs should protect them too.

```
fit_panels(:bacteria)
```



Comparing the four

```
best = minimum(f.aic for f in values(fits))
println("| Step | Mechanism | Parameters | log-likelihood | AIC | ΔAIC |
 ↪  Starts within 0.1 of the best |")
println("|---|---|---|---|---|---|---|")
for (i, mech) in enumerate(mechanisms)
    f = fits[mech]
    @printf("| %d | `%s` | %d | %.1f | %.1f | %.1f | %d of %d |\n", i, mech,
     ↪  length(f.x),
        f.loglik, f.aic, f.aic - best, count(>=(f.loglik - 0.1), f.logliks),
 ↪  length(f.logliks))
end
```

Step	Mechanism	Parameters	log-likelihood	AIC	ΔAIC	Starts within 0.1 of the best
1	`none`	9	-5745.4	11508.8	178.8	8 of 8
2	`decay`	12	-5656.1	11336.3	6.2	8 of 8
3	`applied`	12	-5653.0	11330.0	0.0	7 of 8
4	`bacteria`	12	-5745.4	11514.8	184.8	7 of 8

```
fa, fd, fn = fits[:applied], fits[:decay], fits[:none]
@printf("Steps 2 and 3 improve the log-likelihood over step 1 by %.0f and
 ↪  %.0f units. ",
    fd.loglik - fn.loglik, fa.loglik - fn.loglik)
@printf("In step 3, ι = %.3g copies per day, so untreated frogs sit at %.3g
 ↪  copies; ",
    fa.estimates.ι, fa.estimates.ι / -r0)
```

```
@printf("treated frogs start at %.3g copies on day 0; and Bd varies between
 ↪  swabs with a ",
    fa.estimates.L0)
@printf("standard deviation of %.2f on the natural-log scale (a factor of
 ↪  %.0f). ",
    fa.estimates.σ, exp(fa.estimates.σ))
@printf("*J. lividum* halves every %.1f days on treated frogs and every %.1f
 ↪  days on wild and control frogs.\n",
    fa.estimates.hT, fa.estimates.hW)
```

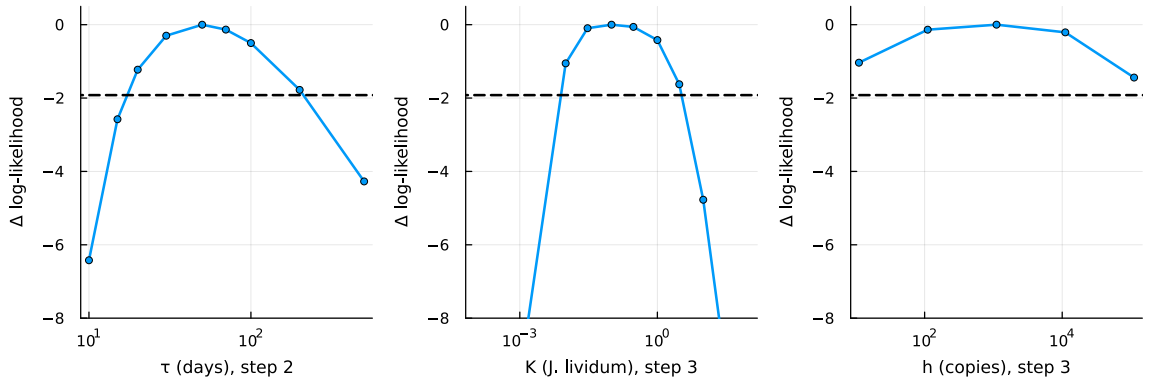
Steps 2 and 3 improve the log-likelihood over step 1 by 89 and 92 units. In step 3, $\iota = 1.15\text{e+}05$ copies per day, so untreated frogs sit at 7.38e+05 copies; treated frogs start at 10.1 copies on day 0; and Bd varies between swabs with a standard deviation of 2.94 on the natural-log scale (a factor of 19). *J. lividum* halves every 4.3 days on treated frogs and every 5.4 days on wild and control frogs.

- Extra clearance on treated frogs is essential. Steps 2 and 3 are far better than step 1, for three more parameters.
- Resident J. lividum does not protect. In step 4 the fit switches the effect off and ends where step 1 does. Controls carried more of the species than treated frogs and had high loads, so the abundance measured by this assay cannot by itself explain the protection.
- The applied bacteria fit slightly better than a waning drug (ΔAIC about 6), but the next two sections show that this preference rests on very little.

How well are the parameters determined?

Profile likelihoods fix one parameter at a time and refit all the others. Values above the dashed line, 1.92 log-likelihood units below the best, form an approximate 95% confidence interval.

```
function profile_plot(fit, name, values; xlabel = string(name))
    pl = profile_likelihood(fit, name, values)
    plot(values, pl .- fit.loglik; xscale = :log10, marker = :circle, ms =
     ↪  3, label = "",
        xlabel = xlabel, ylabel = "Δ log-likelihood", ylim = (-8, 0.5))
    hline!([-1.92]; color = :black, linestyle = :dash, label = "")
end
τs = [10, 15, 20, 30, 50, 70, 100, 200, 500]
Ks = [1e-4, 1e-3, 0.01, 0.03, 0.1, 0.3, 1, 3, 10, 100]
hs = fa.estimates.h .* [0.01, 0.1, 1, 10, 100]
plot(profile_plot(fd, :τ, τs; xlabel = "τ (days), step 2"),
     profile_plot(fa, :K, Ks; xlabel = "K (J. lividum), step 3"),
     profile_plot(fa, :h, hs; xlabel = "h (copies), step 3"); layout = (1,
      ↪  3), size = (900, 300))
```



- Step 2: the suppression wanes on a time scale τ of between roughly 17 and 200 days.
- Step 3: the half-saturation K is between about 0.01 and 3 genomic equivalents, at or below one cell's worth of DNA on a swab. The fit says the bacteria protect fully until they are almost undetectable. The rise in Bd between days 37 and 56 coincides with the last *J. lividum* vanishing, and that coincidence is all step 3 uses.
- The saturating load h is not identified. No group's load passes through the saturating range slowly enough to measure it.

A Bayesian view

Profiles show how far each parameter can move on its own. A Bayesian fit shows what the parameters can do together, and how sure the comparison of mechanisms is. Each parameter gets a weakly informative lognormal prior on its natural scale (`knapp_prior`), with medians at plausible orders of magnitude and spreads covering several of them. r_0 stays fixed. NUTS samples the posterior through the package's Turing extension.

```
using Turing: Turing, NUTS, MCMCThreads, sample, InitFromParams
using MCMCDiagnosticTools: ess, rhat
using Random: Xoshiro
bayes_mechs = (:none, :decay, :applied)
target = (none = 0.9, decay = 0.9, applied = 0.99)   # `applied` diverges at
 ↪  0.9
nchains, ndraws = 4, 500
draws = Dict{Symbol,Matrix{Float64}}()
diagnostics = Dict{Symbol,Any}()
for mech in bayes_mechs
    post = KnappPosterior(models[mech])
    x0 = fits[mech].x
    inits = [InitFromParams((; x = x0 .+ 0.3 .* randn(Xoshiro(i),
 ↪  length(x0)))) for i in 1:nchains]
    chn = sample(Xoshiro(1), turing_model(post), NUTS(ndraws, target[mech]),
 ↪  MCMCThreads(),
        ndraws, nchains; initial_params = inits, progress = false)
```

```
    A = Array(chn)                                  # draw × chain × log
 ↪  parameter
    draws[mech] = reshape(A, :, size(A, 3))
    diagnostics[mech] = (rhat = maximum(rhat(A)), ess = minimum(ess(A)),
        divergences = Int(sum(Array(chn[:numerical_error]))))
end
nothing
```

```
┌ Info: Found initial step size
└   ϵ = 0.05
┌ Info: Found initial step size
└   ϵ = 0.05
┌ Info: Found initial step size
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┌ Info: Found initial step size
└   ϵ = 0.05
```

```
println("| Mechanism | Max R-hat | Min ESS | Divergences |")
println("|---|---|---|---|")
for mech in bayes_mechs
    d = diagnostics[mech]
    @printf("| `%s` | %.3f | %.0f | %d |\n", mech, d.rhat, d.ess,
     ↪  d.divergences)
end
```

Mechanism	Max R-hat	Min ESS	Divergences
none	1.010	1630	0
decay	1.006	971	0
applied	1.006	678	0

An R-hat near 1 and effective sample sizes in the hundreds or more mean the chains agree and the summaries below are stable.

Comparing mechanisms by prediction

PSIS-LOO estimates how well each mechanism predicts a swab left out of the fit (the expected log predictive density, elpd). Unlike AIC, the difference between two mechanisms comes with a standard error, computed from the swab-by-swab differences.

```
loos = Dict(mech => psis_loo(pointwise_loglikelihood(models[mech],
 ↪  draws[mech]))
            for mech in bayes_mechs)
println("| Mechanism | ΔAIC | elpd | Δelpd vs `applied` (SE) | max Pareto k
 ↪  |")
println("|---|---|---|---|---|")
for mech in bayes_mechs
    d = elpd_difference(loos[mech], loos[:applied])
    @printf("| `%s` | %.1f | %.1f | %.1f (%.1f) | %.2f |\n", mech,
     ↪  fits[mech].aic - best,
        loos[mech].elpd, d.difference, d.se, maximum(loos[mech].k))
end
```

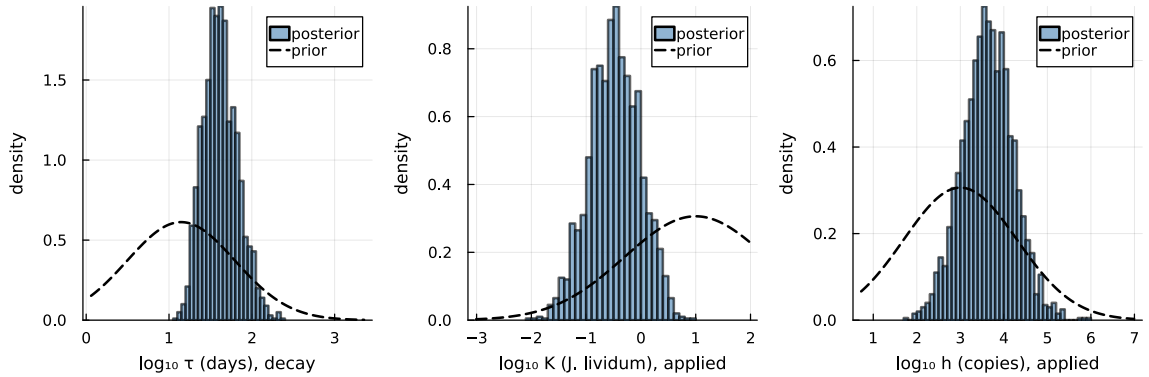
Mechanism	ΔAIC	elpd	Δelpd vs `applied` (SE)	max Pareto k
none	178.8	-5755.1	-89.7 (11.0)	0.28
decay	6.2	-5668.6	-3.2 (2.8)	0.26
applied	0.0	-5665.4	0.0 (0.0)	0.36

- Extra clearance on treated frogs is needed by any measure. `none` loses about 90 elpd, many standard errors.
- **`decay`** and **`applied`** cannot be told apart. The difference is about one standard error. A ΔAIC of 6 looks like a preference, but it rests on a handful of swabs (mostly treated frogs on day 37).
- Pareto k below 0.7 for every swab means the LOO estimates are reliable.

What the posterior says about h, K and τ

```
using Distributions: Normal, pdf
function posterior_panel(mech, name; xlabel = "log₁₀ " * string(name))
    j = findfirst(==(name), parameter_names(models[mech]))
    v = draws[mech][:, j] ./ log(10)
    pr = knapp_prior(models[mech])[name]
    p = histogram(v; normalize = :pdf, bins = 40, color = :steelblue, alpha
 ↪  = 0.6,
        label = "posterior", xlabel = xlabel, ylabel = "density")
    xs = range(minimum(v) - 1, maximum(v) + 1; length = 200)
    plot!(p, xs, pdf.(Normal(pr.μ / log(10), pr.σ / log(10)), xs); color =
     ↪  :black,
        linestyle = :dash, label = "prior")
end

plot(posterior_panel(:decay, :τ; xlabel = "log₁₀ τ (days), decay"),
     posterior_panel(:applied, :K; xlabel = "log₁₀ K (J. lividum),
      ↪  applied"),
     posterior_panel(:applied, :h; xlabel = "log₁₀ h (copies), applied");
     layout = (1, 3), size = (900, 300))
```




```
q(v, p) = sort(v)[clamp(round(Int, p * length(v)), 1, length(v))]
nm = parameter_names(models[:applied])
Xa = draws[:applied]
K = exp.(Xa[:, findfirst(==(:K), nm)])
ratio = exp.(Xa[:, findfirst(==(:Δc), nm)] .- Xa[:, findfirst(==(:h), nm)])
τ = exp.(draws[:decay][:, end])
@printf("Under `applied`, P(K < 1) = %.2f and P(K < 3) = %.2f. ",
 ↪  count(<(1), K) / length(K),
    count(<(3), K) / length(K))
@printf("The clearance rate at low load, Δc/h, is %.0f per day (95%%
 ↪  interval %.0f–%.0f). ",
```

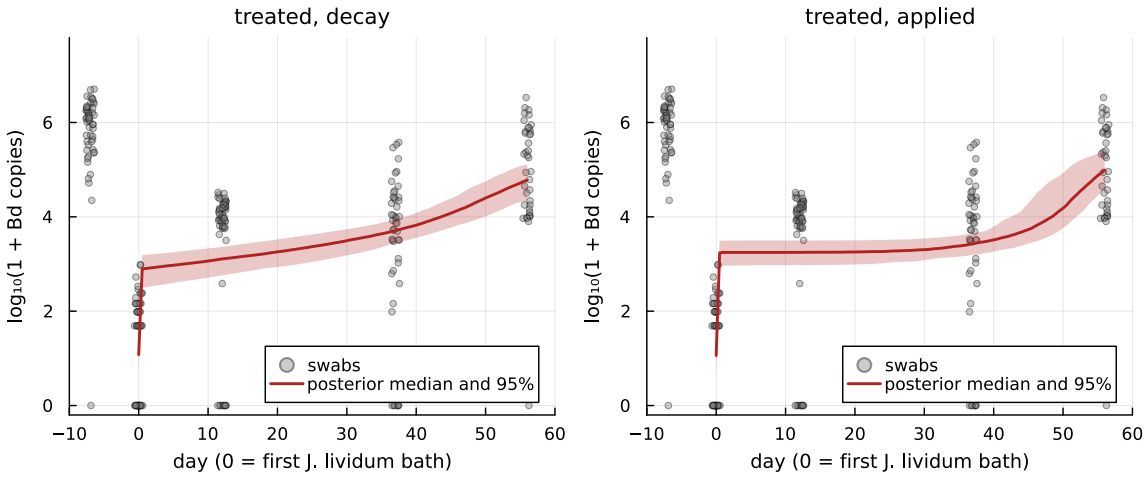
```
    q(ratio, 0.5), q(ratio, 0.025), q(ratio, 0.975))
@printf("Under `decay`, τ is %.0f days (%.0f–%.0f).\n", q(τ, 0.5), q(τ,
 ↪  0.025), q(τ, 0.975))
```

Under `applied`, P(K < 1) = 0.85 and P(K < 3) = 0.99. The clearance rate at low load, Δc/h, is 89 per day (95% interval 39–421). Under `decay`, τ is 41 days (18–114).

- h is weakly identified. It moves together with Δc along a ridge. Their ratio $\Delta c/h$, the clearance rate when loads are low, is well determined. Values of h below a few hundred copies are ruled out, and the prior shapes the upper tail.
- K has an upper bound only. The bacteria must protect until they are nearly undetectable. Below about 0.1 genomic equivalents, the posterior follows the prior.
- τ lies between about 20 and 100 days. That is narrower than the profile interval, partly because the prior (median 14 days) discourages very long waning times and partly because the posterior averages over the other parameters instead of optimising them.

Uncertainty in the fitted loads

```
function band_panel(mech)
    X = draws[mech]
    idx = 1:max(1, size(X, 1) ÷ 200):size(X, 1)
    paths = reduce(hcat, [predicted_loads(models[mech], X[i, :]; days =
 ↪  fine).treated.bd
                          for i in idx])
    keep = fine .>= 0
    lo = [q(log10.(1 .+ paths[t, :]), 0.025) for t in eachindex(fine)]
    hi = [q(log10.(1 .+ paths[t, :]), 0.975) for t in eachindex(fine)]
    md = [q(log10.(1 .+ paths[t, :]), 0.5) for t in eachindex(fine)]
    p = data_panel(:treated; title = "treated, $(mech)")
    plot!(p, fine[keep], md[keep]; ribbon = (md[keep] .- lo[keep], hi[keep]
     ↪  .- md[keep]),
          color = :firebrick, fillalpha = 0.25, label = "posterior median and
 ↪  95%")
end
plot(band_panel(:decay), band_panel(:applied); layout = (1, 2), size = (800,
 ↪  330))
```



The two mechanisms agree at the visits but take different paths between them. `decay` lets loads rise steadily from day 0. `applied` holds them flat until the *J. lividum* is nearly gone, after about day 37, and then lets them rise faster. With swabs only on days 12, 37 and 56, the data cannot tell these paths apart. Swabs are treated as independent here too, so all of these intervals are too narrow if swabs from the same frogs are correlated.

What the fit shows

1. Without an extra effect on treated frogs, the load equation returns them to the wild level within two weeks. Something held their loads down for five to seven weeks after release.
2. Waning suppression from leftover itraconazole and protection by the applied *J. lividum* explain the data about equally well. Every treated frog received both, and no group received only one, so this trial cannot separate them.
3. If *J. lividum* is responsible, it is not because of the species' abundance as measured by this assay. Controls carried more of it and were not protected, so it would have to be a property of the applied isolate, or of how it combines with the drug.

Limits

- Groups, not individuals. Swabs are treated as independent draws. Repeated swabs of the same frogs and the loss of frogs that died both bias the sample. Survivors may carry less Bd than the frogs that died.
- A fixed outside infection rate. ι is constant, although the zoospores in the pond depend on the loads of all frogs. Treating many frogs would lower it. The package's load tipping model includes this feedback; the fit does not.
- Lab parameters. r_0 is taken from Briggs et al. (2010). ITS copies per swab are not sporangia, so ι, h and Δc are on the swab scale and cannot be moved into the sporangium models without a conversion factor.
- A species-level assay. The *J. lividum* qPCR does not distinguish the applied isolate from resident strains.

- Few controls. Controls were rarely recaptured after day 12, and no group was given only itraconazole or only bacteria.

References

- Briggs CJ, Knapp RA, Vredenburg VT (2010) Enzootic and epizootic dynamics of the chytrid fungal pathogen of amphibians. *Proceedings of the National Academy of Sciences* 107:9695–9700. doi:10.1073/pnas.0912886107
- Knapp RA, Joseph MB, Smith TC, Hegeman EE, et al. (2022) Effectiveness of antifungal treatments during chytridiomycosis epizootics in populations of an endangered frog. *PeerJ* 10:e12712. doi:10.7717/peerj.12712